

Fake News Detection using Machine Learning

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Abstract—Fake news shared largely over digital media has become a pressing problem in contemporary information environment impacting on public and producing a misleading effect. The main goal of this project is to work on a model for detecting the fake news using text mining approach for news article. Through-stop-word removal and other techniques of text preprocessing used in the project text data is transformed to features using the TF-IDF method. Algorithms used for classifying the data between fake and real news include decision tree, Logistic Regression, Multinomial Naive Bayes and K-NN. The described model is trained on a labeled data set and evaluated according to the Confusion Matrix. The outcomes prove that the utilization of machine learning makes it possible to create an automatic approach for identifying fake news, which is suitable for large-scale mitigation of the issue.

Index Terms—fake, news, detection, TF-IDF, Logistic Regression, Multinomial Naive Bayes, K-Nearest Neighbor, Decision tree, Confusion matrix.

I. INTRODUCTION

The development of social networks has expanded the availability of information in the field of news already at the international level. However, such ease of access has also hampered the distribution of fake news which is also referred to as fake news. The phenomenon of fake news is generally understood as disinformation or misinformation that are works which spread the adoption of certain ideas or the distortion of the truth. Fake news have negative implications on the society's sanity as it leads to political manipulations, social

radicalization, and public health infections as seen in the recent past.

Because fake news is disseminated on a large scale and at a high speed, attempts to counter the phenomenon tend to remain largely manual. This has given rise to the increased use of automated solutions which are based on machine learning. Fake news detection systems are created to identify and categorize information, trying to mark real news and fake ones based on texts' characteristics and regularities. These systems make use of a variety of text pre-processing methods to get useful information out of the actual text with non-structured text as the input data and converting it to inputs for the learning models.

In this project, an attempt is made to detect fake news from real news using various machine learning algorithms applied on feature engineered articles. In this process, the text data is transformed eliminating preprocessing steps such as stopword removal on the data given as input. Data pre-processing involves feature extraction on the textual data whereby a popular technique known as Term Frequency-Inverse Document Frequency (TF-IDF) is used to transform the textual data to numerical form appropriate for classification. The classification models include Multinomial Naive Bayes, K-Nearest Neighbor and Decision Tree, Logistic Regression, and Support Vector Machine. Performance metrics consisting of accuracy, precision, and recall are assessed concerning confusion matrices for each model.

A. Motivation

The problem of fake news threatens global society today due to the massive increase of misinformation in the social media age. Ideological prejudice due to the actual unrestricted spread of fake news pose a great threat, which can range from political disruption to health risks. That is within this context, there is a presupposition that supports the authority to determine the authenticity of news is vital in providing the public with credible and engaging news. The reason for this project stems from a very vital need of reducing the negative impacts of fake news especially by creating an application that will be able to distinguish fake news. Thus, this project using the general perspective of machine learning advances the goal to develop trustful information and ensure social stability.

B. Objective

In general, the conceptual basis of the work is formulated in the following two primary objectives. First, to create an efficient large scale system for detecting fake news in the stream, which would give a solution to the real world problem of fake news. This then calls for a good efficient and accurate machine learning model that will handle the massive amount of text data needed and classify them appropriately. Second, to view the best methodology that tends to result from comparing the assorted classification algorithms and techniques such as Multinomial Naive Bayes, K-Nearest Neighbor and Decision Tree. From the results of the performance evaluation of these models in terms of accuracy, precision, and recall the project aims at identifying the best approach towards detection of fake news.

II. LITERATURE REVIEW

The slide informs three papers which employ machine learning model for SDC and fake news detection.[1] The first work examines the product reviews in Bengali language. The best performance was recorded by K-Nearest Neighbors with the accuracy reaching only 96.25%. This study contributes to enhance the sentiment detection within a relatively uninvestigated domain. It is helpful in business regarding analysis of customer reviews. But this approach will not turn out to be effective with other datasets of the same nature of data. It can also not process real time data.[2] The second study is on the topic of fake news detection. That is why it presents a new approach based on the K-Nearest Neighbors algorithm fused to the Random Forest model. This model makes accuracy even better to 94%. However, the study does not respond to the problem of falsely selected headlines or real-time detection.

[3] The third study is also for fake news detection. It has Random Forest, it has the highest accuracy of 92.7%. The proposed study enhances the ability to identify misusers across the platforms. It doesn't take into consideration dynamic data source and real-time detection. All the studies indicate that the

various machine learning models have their strong sides. They also identified areas still require further research.

III. METHODOLOGY

In this section the system development and evaluation process used to design this fake news detection system will be described. It accompanied by several main steps of the qualitative process which include data pre-processing, feature extraction, and use of diverse machine learning techniques. All of them are in a way geared towards improvement of accuracy as well as efficiency because the ultimate goal of the model is to classify news articles as fake or real. The next subtopics specify the means employed to achieve goals in every phase of the work..

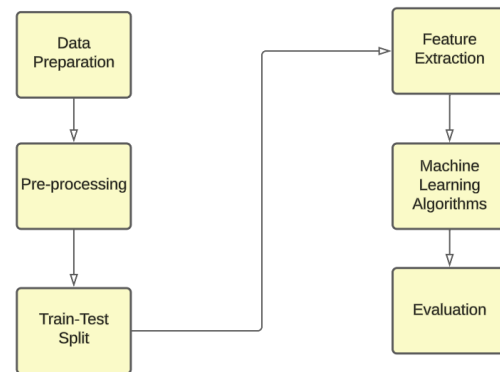


fig-01: Representation of the methodology

A. Data Preparation

The first process in the project is data pre-processing. The dataset, collected from Kaggle, consists of two separate files: one of them filled with fake news and the other one with real news articles. In order to make it easier for the machine learning model to distinguish the two types of news, a new column called 'Class' was created in both newscasts. This column in the fake news dataset is given a '0' to label all entries as fake while the one in the true news dataset is given a '1' to label all entries as real news.

Subsequently, in both datasets, all other features that are not the text of the news articles and the "Class" label were eliminated if they appeared to be redundant. Afterwards, the two sets of data were joined into single set of data as the next step in data analysis. In order to avoid instructing the model to learn from the order columns were enter to be fake or real news, the merged dataset was randomly shuffled equally between fake and real news articles.

This step actually helps to acquire a well labeled dataset free from any mess as well as noise that can affect the models' ability of identifying fake news.

B. Pre-processing

The following step after preparing the dataset is text pre-processing so that the data can be in a right format required by the machine learning algorithms. Text data can come in different cases and has special characters and words that are irrelevant to the model. To overcome this, several preprocessing techniques were applied.

First, all alphabetic characters in the news articles were lowcased to avoid computational complexity in matching. This removes discrepancy between words that are the same, but are only different in capitalization hence, the words “Fake” and “fake”.

After that, all unnecessary symbols or signs, as well as empty words such as URLs (www, https, com, org) were deleted from the text. This assists in the cleaning process of the data in order to remove noise that does not add any semantically related meaning to the news articles. These symbols and terms are familiar in reading news articles but lack accurate meanings as to decide whether a news article is real or fake.

This I have found to be an important step in data cleaning wherein noisy data is removed from the dataset so that the machine learning algorithm is able to extract features from the text and make accurate classifications from the actual textual content.

C. Train-test split

In this step, the dataset is divided into two parts: Among them, there is a data set for training and another one for assessment of the models’ effectiveness. Once again the 80%-20% division was adopted so that eighty percent of the samples was used to train the models and twenty percent was reserved for testing.

Training dataset is used here to develop and train the machine learning algorithms so that the algorithm gets to understand relationships existing in between the text features and ‘Class’ labels which can be either fake or real news. In this case, the test dataset which the model has not been trained on will be used to determine the level of generalization the model. This partition ensures independent check on the ability of the model to learn and generalize from the data without over complicating the program to give high accuracy on the training data but poor performance on unseen data such as new news articles.

This division is typical for machine learning: still, enough training examples are needed, while still the model should be tested on unseen examples.

D. Feature Extraction

This is actually the step of getting further preprocessing on the kind of data that the word2vec specializes in which is the raw text data. Because of this, raw text data cannot be fed into machine learning algorithms in its raw form because they take in numerical data. To eliminate this gap, what is called Term Frequency-Inverse Document Frequency (TF-IDF) vectorization was applied. TF-IDF is among the most adopted methodologies in text-based application in the machine learning approach, such as fake news identification.

The TF-IDF vectorizer works on the basis of two factors; the first is a measure of how frequently a particular word appears in a document. The Term Frequency (TF) component shows the frequency of occurrence of a specific word inside the particular article as the indicator of the importance of this term inside the given article. The number (TF) of any word in a document increases as the word appears more frequently in the document. However, not all of the most recurrent words are significant for classification as most of them are stop-words: “the”, “is”, “and” can be met in virtually every text. On the other hand, to avoid such an effect, the Inverse Document Frequency (IDF) component determines the rarity of the word in the entire corpus. Thus, frequently appearing words in many articles assigned low IDF scores, whereas rare, yet informative words assigned higher IDF scores.

TF-IDF is achieved by multiplying the TF values by the IDF values; this makes it place most importance on words that recur often within a particular article and are however rare in the whole corpus. For instance, in the case of fake news identification, words such as “hoax” or “conspiracy” will be quite common in fake news but insignificant in actual news. These words will receive higher TF-IDF scores which will cause machine learning to concentrate on the most informative features

Previous work related to text document processing results in the matrix containing the vectors, where each vector represents news articles, and each element of the vector is a numerical value that works as an index of TF IDF words. These vectors are used in the next step of the machine learning model where the light control field of Deep Learning is used to distinguish whether they are genuine news or fake news based on the patterns formed in the texts.

E. Machine Learning Algorithms

The second step that follows text data pre-processing is the use of machine learning algorithms for the classification of the News articles as Real and Fake. For this project, multiple algorithms were employed to compare their effectiveness in detecting fake news: Some of the prominent algorithms some

of which were used for the analysis include: Logistic Regression, Decision Tree (DT), Multinomial Naïve Bayes (MNB), and K-Nearest Neighbor (KNN).

Logistic Regression was used as a classification technique that estimates the probability of the binary outcome using one or more features (*, the TF-IDF scores of the words). It does this by modeling the data by a curve of logistic function to enable one estimate the likelihood that given news story belongs to either the real or fake class. Logistic Regression is easy to use and works efficiently for the most common database problem type – binary classification and is linearly soluble with readable results.

The other model used in the classification task was the Decision Tree (DT) model. Decision Trees are a structure in which, given certain analysis conditions (nodes), the data is divided based on the features' values (words or their TF-IDF scores). The goal is to form a tree structure that results in a decision, whether real or fiction, that has little or no classification errors. Another common form of splitting a dataset is the Decision Trees which are also easy to use, can handle relationships of high degree and are easy to interpret, however they are more sensitive to noise within the data they are dealt with.

This is why the Multinomial Naive Bayes (MNB) algorithm works best for the text classification where data is counted in word frequency or probability. It works on the basis that all the features or words are independent of each other if the class is the real or fake news. This assumption is sometimes violated in real application, but Naive Bayes is considered as simple and fast learning algorithm that works good on the textual data. In this project, MNB was differentiate the chances of a news article being fake or real according to the words contained in it.

Last, the K-Nearest Neighbor (KNN) creates a model using which a new point (an article) will be compared to the points of the training set. This uses 'k' nearest neighbor approach, which considers the point in a given proximity of 'k' known distance measure such as Euclidean distance then the point is classified depending on the class of the closest point. Although KNN is easy to understand and implement the outcome is sensitive to the choice of 'k' and the distribution of the input data. In this project the KNN was used to assess the performance when applied to text data even though KNN may not be relevant in high dimensions vectors like the TF-IDF.

Every algorithm was trained on the prepared dataset, and the definite results depending on the metric were obtained to characterize the accuracy of classification and measure the precision, sensitivity, and F1-score of every algorithm. Using cross- validation with several classification algorithms, the given project intends to recognize which out of all algorithms proposed result to the highest precise fake news detection. Then, compare the advantages and disadvantages of each algorithm.

F. Evaluation

Subsequently at the final stage of the methodology the effectiveness of the machine learning models is tested through Confusion Matrix. The classification results in this tool are supported by the model outputs where one can determine the accuracy of fake and real news. Confusion matrix given in Figure 11 shows the summary of the performance in terms of actual and predicted class labels.

In this project, the confusion matrix categorizes predictions into four key metrics: Sensitivity or True Positive rate (TP), Specificity or True Negative rate (TN), Over Prediction (FP), Under Prediction (FN). True Positives measuring genuine news articles that have been classified as real, True Negatives are fake news articles to be categorized as fake. False Positives are invented when a fake news article is accepted to be a real news article, and False Negatives are viewed when a real news article ends up being classified as fake.

From the confusion matrix, several evaluation measures can be estimated common of which are accuracy, precision, recall, F1-score, etc. Accuracy gives the overall estimation of the model while precision shows the efficiency of the model in terms of the presence of actual/hard positive cases among all the cases it suggested as positive ones. Recall checks the potential of the model to identify all the positive-class patterns correctly, whereas F1-score creates a combination of metrics between both precision and recall.

Studying the confusion matrix and these indicators, the project defines the advantages and disadvantages of each of the algorithms used in the fake news detection system, and thereby selecting the most appropriate model for practical application.

IV. RESULT ANALYSIS

By analyzing the performance metrics achieved by the different machine learning algorithms used in the fake news detection project I have noticed that they differ a lot. By measuring the accuracy of each algorithm as well as false positives and false negatives, their performance with regard to the classification of news articles is best understood.

Consequently, the performance of Logistic Regression was quite outstanding with an accuracy of 98.78% for the identification of genuine and fakes news. Again, there were cases where actual fake news articles were classified as real; 48 out of the news articles were classified as fake though they are real, and 62 articles of news that were classified as real but in real sense they are fake. These outcomes confirm the model's high reliability, but demonstrate that false negatives can be eliminated; however, further research is needed to improve this model.

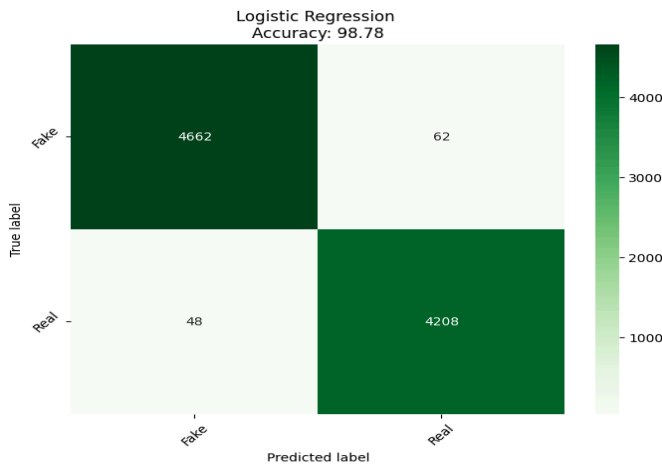


fig-02: Confusion matrix of Logistic Regression

Thus, the Decision Tree algorithm turned to be even more accurate as it had the highest accuracy of the 99.64% among all the models used in this work. It identified the fake news wrong with only 23 articles being real and 9 forecasted to be fake when they were real. This a very low misclassification rate and shows that complicated relations within the data can be captured by the Decision Tree method well enough to be used on real data for fake news detection.

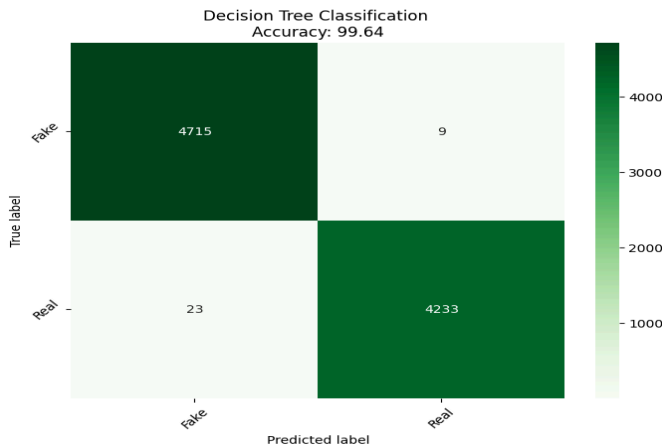


fig-03: Confusion matrix of Decision tree

The next model called the Multinomial Naive Bayes algorithm provided slightly lower accuracy of 93.65%. It was massively imprecise, misidentifying 314 articles as fakes when they were actually real articles; and 256 articles, it labeled as genuine articles when in actual sense, they are fake articles. These results demonstrate that while MNB has applicability in specific contexts, its performance may be suboptimal in the context of fake news detection, which has to be attributed to MNB's independence assumption.

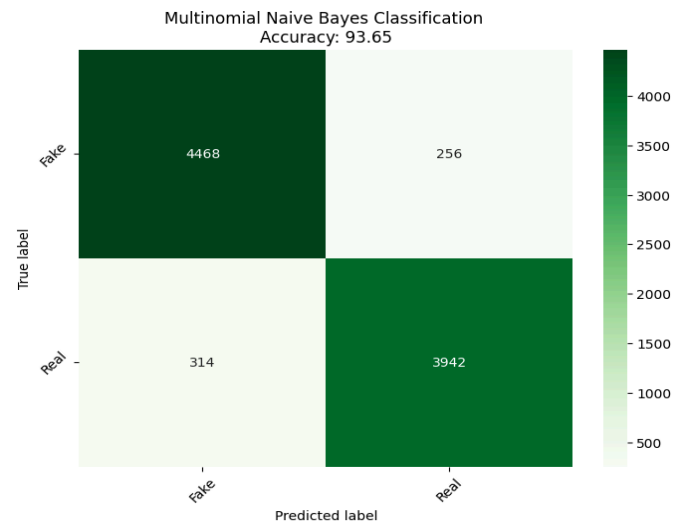


fig-04: Confusion matrix of Multinomial NB

Last of all, the results shown that the K-Nearest Neighbor (KNN) algorithm scenario was only 70.18% accurate. In further detail, this model set up 2565 articles to be fake while they are real news; 113 articles were also set up to be real when they are fake news. A relatively high number of misclassification suggests that KNN can hardly be the best choice to be applied to this high-dimensional text data classification problem since the algorithm usually degrades fairly when there is much noise and irrelevant features.

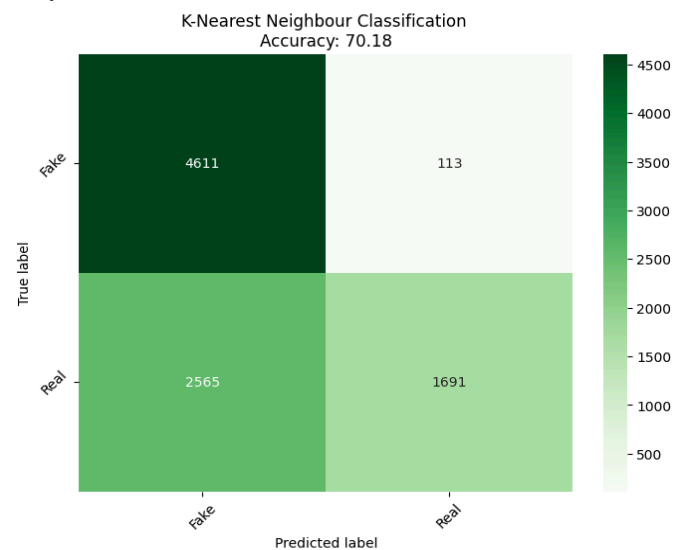


fig-05: Confusion matrix of KNN

In summary, the present evaluation profoundly indicates that the Decision Tree and Logistic Regression algorithms dominated other models thereby, rendering them as the best models suitable for developing and implementing in a possible real-time fake news identification system.

V. CONCLUSION

Finally, this work showed the applicability of different machine learning techniques for the fake news classification problem, which is essential to combat false information in

today's society. Performing the experiments, we estimated various methods including Data preparation, Preprocessing the Document, and creating TF-IDF Features on models such as Logistic Regression, Decision Tree, Multinomial Naive Bayes, and K-Nearest Neighbors. The Decision Tree and the Logistic Regression showed the highest accuracy, 99.64% and 98.78%, respectively which reveal the efficiency of the classifier in the context of the news article classification as well as the minimum error rate of the method. While using inaccurate and possessing high misclassification rates, the performance of Multinomial Naive Bayes and KNN were lesser than the two spoken above. These results stress the necessity of proper algorithm choosing regarding the solution of particular problems and recurrent enhancement of model quality. As part of the effort to fight fake news, this project is beneficial and would open the possibility to further develop new goals within machine learning for immediate detection systems.

VI. REFERENCES

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